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A Review on Upper-Limb Myoelectric Prosthetic Control

Nisheena V. Iqbal¹, Kamalraj Subramaniam² and Shaniba Asmi P.¹

¹Department of Electronics and Communication Engineering, Karpagam university, Coimbatore, India; ²Department of Electronics and Communication Engineering, Karpagam University, Coimbatore, India

ABSTRACT

One of the most interesting topics in the field of rehabilitation is that of upper-limb myoelectric prosthetic control. It is a technique by which prostheses are controlled by means of surface electromyogram (sEMG) signals collected from remnant muscle tissues at the residual limb of an amputee. Intuitive control of multifunctional upper-limb prosthesis can be accomplished using pattern recognition (PR) of sEMG signals. In spite of the tremendous progress made in the research of the so-called mind-controlled artificial arm, none of the academic achievements has yet reached the end users. This review paper portrays the current state-of-the-art approach in sEMG pattern classification-based control, identifies the factors that hinder the clinical usability of the system and focuses on the recent research directions toward translating the academic findings into a commercially acceptable robust myoelectric prosthesis. Control strategies proposed for simultaneous and proportional control (SPC) of multiple degrees of freedom (DoFs), which is identified as the most significant barrier for the transition from laboratory to clinical practice, are discussed. Directions for future research are also briefly outlined.

KEYWORDS

Classification; Degrees of freedom (DoFs); Feature extraction; Myoelectric; Electromyogram (EMG); Prosthetic control; Pattern recognition; Simultaneous and proportional control; Upper-limb amputation

1. INTRODUCTION

Though the proposal for prosthesis based on myoelectric control (MEC) dates back to 1940s, due to limitations in technology a substantial progress in the area had occurred only in the late 1960s. In this version, which is still used by most of the upper-limb prostheses available commercially, the device is operated either in on/off or proportional mode by comparing surface electromyogram (sEMG) amplitude to a threshold. But here the number of reliable functions that can be controlled per channel was limited to three and the users found it very hard to master the muscle contraction procedure [1,2]. These disabled people had always been expecting “a feeling of self” with respect to the prosthesis.

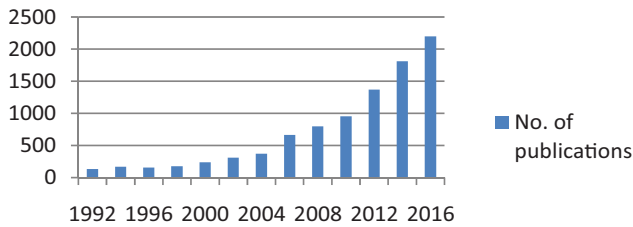
The emergence of pattern recognition (PR)-based control was a significant milestone in the development of myoelectric prostheses. Here it was assumed that each of the intended movements can be mapped to a specific sEMG pattern. The major steps consist of performing sEMG measurement, extracting distinguishing features of the captured sEMG patterns, classifying different intended movements and commanding the prostheses controller to implement the action. Last few decades witnessed remarkable improvement in the classification efficiency of these PR-based algorithms, utilizing different types of sEMG features and classifiers to recognize the user intent. Thus, PR-based control emerged as the most promising method

to operate the upper-limb prostheses intuitively and easily. Moreover, innovative surgical procedures such as targeted muscle reinnervation (TMR) provided physiologically appropriate myoelectric signals (MES) associated with the former functions of the missing arm. This helped in using PR control even in above elbow amputees [3]. The remarkable increase in upper-limb prosthetic research can be seen from Chart 1; based on a simple online search with the term myoelectric upper-limb prosthetic control.

Although PR control for myoelectric upper-limb prostheses achieved classification accuracies greater than 95%, none of them has been available commercially till date. The only one prosthesis based on pattern classification method released to the market toward the end of 2014, Coapt's complete control system, however, could only effectively meet the needs of patients who underwent TMR surgery [4,5]. This reasserted the finding that classification accuracy did not automatically lead to good usability of a controller. Thus, the significance of improving classification accuracy in PR-based MEC dropped gradually. A fairly extensive review on upper-limb prosthesis users and the requirements of the prosthetic system for better acceptability and usefulness was given by Cordella et al. [6].

Inability to control multiple DoFs in a simultaneous and proportional manner, lack of sensory feedback in normal

task performance, absence of adaptive signal processing algorithms to mitigate the effect of muscle contraction variation, limb position changes, electrode location changes, muscle fatigue, etc., and the lack of sensor fusion approaches are the factors that inhibits the development of PR control into a clinically viable system. To bridge this gap between state-of-the-art approach in research and market, the focus of research need be shifted from increasing the classification accuracy to mitigating the effect of these factors [7].



This paper is organized as follows. Section 2 gives a detailed review on different feature extraction and classification methodologies used in PR-based myoelectric control of upper-limb prostheses. Section 3 explains the factors that limit the development of the PR-based system into a commercial product even after achieving classification accuracy above 95%. Section 4 gives a detailed review about the methods for achieving SPC of multiple DoFs, one of the main factors that widens the gap; and determines encouraging research that may bring desirable improvements in the control, and finally conclusions are drawn in Section 5. Among the current approaches for MEC, SPC of multiple DoFs, which is the most essential feature to be possessed by a commercial prosthesis, is more likely to be achieved via regression based approaches which have recently given encouraging results [8–11]. Newer regression algorithms combining the best attributes of PR control could be the key to more robust and reliable control.

2. MYOELECTRIC CONTROL BASED ON PATTERN RECOGNITION

Initially myoelectric prosthesis was controlled primarily by the estimate of amplitude or the rate of change of sEMG signals [1,2,12]. Although these direct control systems achieved high success rate, only one or two degree of freedom (DoF) could be controlled at a time. But control of multiple DoFs is desirable, especially for high level amputation. PR turned up as a solution to overcome this limitation of conventional control. These systems operate on the assumption that the MES pattern will be repeatable for a given movement class, and distinct for different movement classes [2]. Classification accuracy is the main metric to

evaluate the performance of a PR-based system. Classification accuracy remained almost the same irrespective of the type of EMG, whether intramuscular EMG (iEMG) or sEMG. Moreover, unlike iEMG, sEMG is non-invasive, which makes it the right candidate for PR-based myoelectric prosthetic control. Signal detection, preprocessing, data windowing, feature extraction, dimensionality reduction, classification, post-processing and proportional control are the different steps involved in PR control.

Initially, only a few input channels were used to detect a signal because the technology of the day was not matured enough to measure and process multichannel inputs. But as the technology advanced over years, real time processing of data from multiple channels could be done with less complexity. However, the number of channels is limited by the feasibility of embedding the electrodes in the socket of a prosthetic limb and the number of control sites available after amputation. The second limitation could be overcome by innovative surgical methods like TMR [3].

Features are extracted from analysis windows that are formed by segmenting the MES data. There is a tradeoff between the length of analysis window and the processing time required to reach at a decision. The window should be short enough to avoid long processing delays and long enough to make a good decision. For a reliable system maximum allowable delay between signal generation and prosthetic actuation must be less than 300 ms [13]. Window length is usually taken as 100 to 250 ms. The windows may be either disjoint or overlapped. Overlapping analysis window that is capable of producing a dense decision stream, with regard to the available computing capability is generally preferred. Features extracted from these windows describe the pattern associated with each limb movement. Many different features based on time domain (TD), frequency domain and time-frequency domain methods have been proposed in the literature. Once the features are extracted, dimensionality reduction techniques are generally employed to reduce the size of the high dimensional feature space. Various strategies for feature subset selection and feature combination are available in the literature to meet the purpose.

A classifier is used to discriminate the intended movements from the extracted features. Numerous types of statistical and learning classifiers have been studied and analyzed for myoelectric control [14–16]. However, it was proved that, if an appropriate set of features is available, almost all classifiers will perform similarly [14,17]. Thus, more emphasis shall be given to choose appropriate

features rather than choosing a classifier. Post-processing can be done by majority voting algorithm. This provides smooth transition between motions and removes misclassification in making decisions. The speed of DoFs is controlled by proportional control. In earlier systems where there was a one to one mapping of the channel to the DOFs, this was almost straight forward and easily feasible [18]. But in PR systems the task was more challenging.

2.1 Feature Extraction

Choice of a feature set plays significant role on the performance of a PR system. Non-parametric, parametric and combination of both have been used in myoelectric prosthetic control. These feature sets with classifiers used are shown respectively in Table 1, Table 2 and Table 3. In 1960s and early 1970s features based on MES amplitude were made use of which achieved an accuracy of about 75%. A large number of electrodes were used in these systems which made it cumbersome. Parametric features using auto regressive (AR) models or auto regressive moving average (ARMA) models required only few channels and could discriminate between muscle activation patterns even for the same myoelectric power levels

[19–22]. Sequential least-squares (SLS) algorithm used to generate time series AR parameters possessed large computational overhead [19].

The same parameters could be generated by using Hopfield network in a computationally efficient way [23]. Kang et al. [24] modified AR coefficients to get the cepstral coefficients whose mean absolute values (MAVs) were found to be at least 5% superior to AR coefficients.

In 1993, Hudgins et al. proved that sEMG signals exhibited a deterministic structure during the onset of a muscle contraction [13]. Feature set consisting of non-parametric features like MAV, MAV slope, zero crossings (ZC), slope sign changes (SSCs) and waveform length (WL) were extracted from the transient MES and could achieve a classification accuracy of roughly 90% in a four class problem when classified with artificial neural network (ANN). An embedded control system was also implemented on the basis of this approach [25]. These features were later come to be called as Hudgin's TD features. As transient MES patterns had structures in both time and frequency, it could be expected that signal energy would be best concentrated in a dual

Table 1: Non-parametric features

Features	Classifiers	Subject, channels	Pre-/post-processing	Refs.
Hudgin's TD feature set	ANN	4N+1A; 1ch		[13]
	LDA and MLP	16N; 4ch; 2cl	PCA feature projection	[39]
	GA	5A; 4ch; 6cl		[25]
	Fuzzy classifier	4N+1A; 1ch		[13]
	LDA	12N; 4ch; 4cl	Majority voting	[43]
	GA	4N+1A; 1ch		[13]
	GMM, LDA, LPN, MLM	12N; 2ch; 6cl	Majority voting	[14]
	SVM, LDA, MLP	11N; 4ch; 6cl	Majority voting	[45]
	LDA, MLP	16N; 2ch; 4cl	PCA feature projection	[39]
	LDA	11N; 4ch; 6cl	PCA dimensionality reduction	[28]
STFT, WT, WPT	MLP-NN	4ch	Linear-nonlinear feature projection composed of PCA and SOFM	[33]
WPT	NN	10N; 2 ch; 4cl		[57]
WPT, AR coefficients	MLP	10N; 4ch; 9cl	LDA performs dimensionality reduction and clustering of WPT features.	[44]
DFT (mean and variance values)	Modified KNN	4N; 2ch; 30cl		[41]
Short-time Thomson transform	NN	1N; 32ch; 8cl		[36]
FD and MFL	ANN	5N; 4 ch; 4 cl		[34]
FD and MFL	TSVM	7N;2ch; 4cl		[45]
ICA	TSVM	7N;4ch; 4cl		[46]
DFA	LDA	20N;5ch; 8cl		[35]
RMS	TSVM	8N+1A;4ch;16cl		[47]

A – amputee; ch – channel; cl – class; N – normally limbed person.

Table 2: Parametric features

Features	Classifiers	Subject, channels	Pre-/post-processing	Refs.
Eleventh-order AR coefficients	Modified KNN	4N;2ch; 30 cl	GA subset selection	[41]
Sixth-order AR coefficients, RMS value	Single mixture GMM	11N; 4ch	Majority voting	[42]
AR coefficients	Fuzzy clustering NN, MLP, conic section function NN	1H; 1ch; 6cl		[40]
Sixth-order AR coefficients, RMS value	HMM,MLP	11N; 4ch; 6cl	Majority voting	[32]
Sixth-order AR coefficients, RMS value	GMM, LDA, MLM NN and LPN	12N; 4ch; 6cl	Majority voting	[14]
second-order AR coefficients, RMS value	SVM, MLP	11N; 4ch; 6cl	Majority voting	[45]
Cepstral coefficients derived from AR coefficients	Euclidean distance measure, weighted distance measure and maximum-likelihood method	6N; 2ch; 10cl		
AR coefft, IAV, linear cepstrum coefficient, adaptive cepstrum vector	Artificial intelligence	6N; 1ch; 6cl		[38]

Table 3: Combination of parametric and non-parametric features

Features	Classifiers	Subject, channels	Pre-/post-processing	Refs.
Sixth-order AR+RMS +Hudgin's TD	GMM	12N; 4ch; 6cl	Majority voting	[14]
Sixth-order AR+RMS+Hudgin's TD+other sets	SVM,LDA, MLP	11N; 4ch; 6cl	Majority voting	[45]
TDAR	Multiple binary classifier, LDA	12N; 7cl		[58]
Hudgins TD + 5 additional features	GA	5A; 4ch; 6cl		[25]
Variance value	ANN	5n; 4ch; 6cl		[7]
MAV+WA+VAR+WL	Feed forward MLP	5N; 1A; 64ch; 12cl		[49]

representation like time-frequency domain [26]. Thus, short-time Fourier transform (STFT), wavelet transform (WT) and wavelet packet transform (WPT) outperformed Hudgin's TD features with accuracy about 94% in the same four class problem.

When transient MES was used, switching from class to class was not feasible in an effective and intuitive manner since the control input needed the contraction to be initiated from rest each time. The same feature sets, Hudgin's TD feature set and time frequency feature set could be used with a continuous classifier [27]. Time frequency features which were very powerful in dealing with transient MES could not perform as well as Hudgin's TD features in dealing with continuous MES [28]. To increase the classification accuracy an optimum wavelength decomposition method was tried to select an optimal basis from the library of wavelet packet bases for the continuous classification scheme [29,30]. Symlet wavelet was proved to be the optimal one using the method.

Though very high accuracies could be achieved when using different feature sets, research continued in increasing the classification accuracy using new feature sets and combination of existing feature sets [31]. The relative performance of three feature sets, Hudgin's TD features, AR+ root mean square (RMS) features and AR+RMS+TD features was compared [32]. AR+RMS+TD feature showed better classification accuracy than the rest of the feature set. But this was computationally expensive as it outputs a relatively large feature vector. A direct comparison of feature sets, Fourier transform (FT), WT, TD, stationary WT, WPT and AR coefficients was done by Hargrove et al.; sixth-order AR coefficient exhibited the highest classification accuracy. A comparison was also made between surface and intramuscular inputs using the features, Hudgin's TD, sixth-order AR coefficients+ RMS signal values and concatenated TD and sixth-order AR coefficient feature sets[TDAR]. TDAR feature set outperformed all others but the slight improvement over the TD features was achieved at the expense of considerable processing overhead [14,17]. Oskoei et al. considered the performance of wide range of features individually and in groups [33]. The study

involved individual features of the TD such as MAV, RMS, WL, VAR, ZCs, SSC, Willison Amplitude (WAMP) and two types of modified MAVs. Individual frequency domain features considered were power spectrum, AR coefficients of order two and six, the mean and median of signal frequencies. In addition to these features, multivariate feature sets like Hudgin's TD features, RMS + sixth-order AR coefficients; MAV+WL and RMS + second-order AR coefficients were also dealt with. Hudgin's TD features outperformed all other features due to its high accuracy rate, stability against varying segment length, low alteration over repeated sessions and computational simplicity [33]. Based on the findings of Gupta et al., sEMG signal possesses fractal properties and the experimental study conducted by Arjunan et al. concluded that combination of fractal dimension (FD) and maximum fractal length (MFL) features of single channel sEMG can reliably identify movements, even when muscle activity is very weak and multiple muscles are involved [34]. Another fractal feature, detrended fluctuation analysis (DFA), exhibited even better discrimination accuracy, particularly for low level and equal power sEMG signals [35]. The use of fractal analysis may be suitable for amputees who are unable to perform high level muscle activations and to make a significant difference in power between movements.

2.2 Classification

Primary goal of a classifier is to identify the patterns and to classify them. Diverse types of linear and nonlinear classifiers have been tried for MEC. Conventional PR technique based on time series parameters achieved good class separability, but at the expense of excessive subject training and slow implementation [19]. The application of neural networks as classifiers turned out to be a solution to both these problems. A two layer perceptron with back propagation algorithm was used to classify a single site myoelectric signal based on two features, the first time series parameters and the signal power. Though the momentous potential of neural networks for application of MES analysis was exposed, these single channel systems were computationally complex and were just confined to laboratory implementation. However, better performance was obtained on using

ANN along with Hudgin's features [13]. Kang et al. made a comparison between linear classifiers based on Euclidean distance measure (EDM), weighted distance measure (WDM) and maximum-likelihood method (MLM) [24]. WDM which was highly successful in speech processing proved unsuitable here, because unlike speech signals, the standard deviation for features with the same index is not similar for MES. Though MLM performed well with cepstral coefficients, it possessed the highest computational overhead among the three.

Park et al. proposed a method based on genetic algorithm (GA) which could do better identification of motion classes with less computing time and little subject training [36,37]. An evidence accumulation process was carried out for PR where a fuzzy mapping function was used to transform EDM. Jangwoo et al. applied GA to hybrid markov model (HMM)-multi layer perceptron (MLP) hybrid and HMM-counter propagation network (CPN) hybrid classifiers [38]. By generating proper initial connection weights GA could speed up the learning time of the suggested models. Even though the hybrid HMM-GA-CPN classifier gave more rapid error convergence rates, the recognition rates of HMM-GA-MLP was found to be far superior. When using ANN, which was proved to be a good candidate for MES classification, the dynamic aspects of sEMG signals were not taken care of. Though HMM could not show a superior performance when applied to sEMG, discrimination, it is known to be an excellent tool for handling the dynamic behavior. The benefits of the two methods accounted for the superiority of HMM-MLP hybrid [38].

It was shown that the performance of linear discriminant analysis (LDA) is in par with MLP when the feature set is preprocessed with principal component analysis (PCA) [26]. Unlike LDA, MLP has the capability to prescribe non-linear class boundaries, but LDA is much easier to implement, faster to train and does not compromise classification accuracy. PCA projects the high dimensional feature set into a relatively low dimensional space, while preserving the linearity. This made LDA superior to MLP.

A fuzzy classifier also outperformed ANN when applied on the pattern classification method used by Hudgins [13,39,40] on transient sEMG signals. The primary advantage of fuzzy system was that patterns in data which were difficult to be detected by other methods could be easily discovered. They also proved Gaussian mixture model (GMM) to be an effective method on continuous sEMG signals [41]. Though an increase in

the number of mixtures could increase the accuracy of modeling, larger computational cost associated with additional mixtures and the lack of sufficient training data made the single mixture GMM an effective classification methodology. The superior performance of GMM was reiterated by Huang et al. [42] in his comparative study. The average GMM performance would increase if optimum mixture number was selected for each subject. Karlik et al. [43] made a comparative study on the classifiers: MLP-NN using back propagation, conic section function NN (CSF-NN) and fuzzy clustering NN (FC-NN). More reliable and accurate classification rates (98.3%) were obtained with FC-NN. Though good classification accuracy was produced by MLP-NN, training time required was approximately double of that required by FC-NN. CSF-NN was also fast, but the classification accuracy was less when compared with the other two. It was proved that HMM also outperformed MLP when computationally inexpensive methods were used to determine HMM parameters [42]. The low complexity of HMM in its operational mode and training mode, made it a suitable candidate for real time implementation and adaptive classifier training. Arjunan et al. used the modified SVM developed by Jayadeva et al. [44], which has been named as twin SVM (TSVM) to classify fractal features. TSVM outperformed the standard classifiers which used error minimization based learning methods, when used with fractal features [45] and ICA separated sEMG [46], to classify hand movements. In ICA-separated sEMG, individual motor units are treated as independent sources for isolating individual motor unit action potentials from sEMG. The activity from an entire muscle is accounted by the superposition of the activities from each of the motor units within the muscle. The effectiveness of TSVM in multi-class prosthetic control with unbalanced dataset was reiterated by Soman et al., when used even with a simpler feature, RMS value [47].

2.3 Feature Extraction vs. Classification

A direct comparison was made by Hargrove et al. among the classifiers: MLP, LP, LDA, GMM and HMM [17,48]. Data was collected from 12 normally limbed subjects by placing four equally spaced electrodes about the circumference of the forearm for six classes of movements. Figure 1 (a) shows direct comparison of various feature sets made. In Figure 1 (b), it was illustrated that, for a given feature set all the classifiers gave almost same classification accuracy. Since feature representation was found to be more influential than classifier selection [14,17], classification test could be thought of as

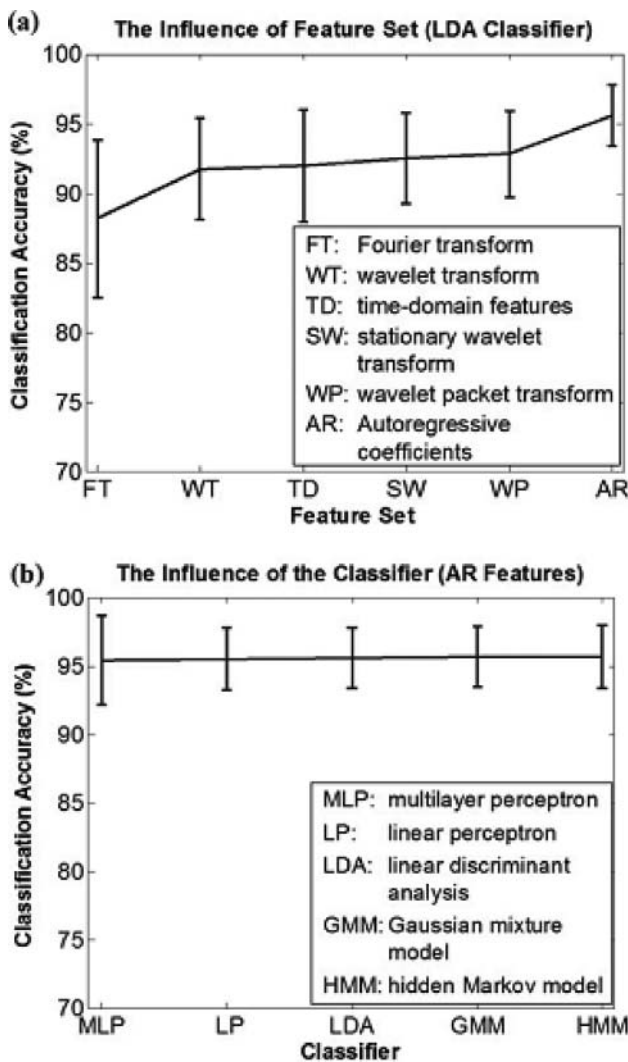


Figure 1: Effect of feature set and classifiers on classification accuracy [17]

Source: Reprinted with permission of IEEE from: L. J. Hargrove, K. Englehart and B. Hudgins, "A Comparison of Surface and Intramuscular Myoelectric Signal Classification," *IEEE Trans. on Biomed. Eng.*, vol. 54, no. 5, pp. 847- 853, May 2007.

essentially a linear problem if an appropriate feature set was present.

More complex and powerful classifiers may be constructed, but it has been shown that LDA classifier does not compromise classification accuracy, is simpler to implement and is faster to train. Now, it is mostly accepted that the combination of TD features with Fischer LDA provides a good balance between algorithm performance and computational efficiency [7].

3. GAP BETWEEN ACADEMIA AND INDUSTRIAL STATE OF THE ART

In the last few decades tremendous amount of research has been carried out in the area of myoelectric prosthetic

control, mainly focused on PR-based algorithms. But none of these research achievements could be implemented in commercial systems. Focus of the research carried out so far in this area was mainly limited to improve the classification accuracy in laboratory off line tests that were quite contrary to the real life scenario of the user. This accounts for the wide gap between academia and the commercial state-of-the-art in this area [4,7,9,17,48–50]. It is more appropriate to put the user at the center of research and shape research to target clinically relevant outcomes rather than increasing the classification accuracy. The factors that lead to this disparity is discussed below.

Most of the works on MEC were done in ideal laboratory conditions. They did not accommodate the changes in sEMG patterns that usually arise in real life situations, due to physical and physiological variations such as muscle fatigue, electrode shift, contraction intensity changes, arm position changes [17,51–54], etc. This accounts for large variations in the user's ability to ensure an efficient prosthetic control especially for long durations [7]. Incorrect operation of the prosthesis that results from these variations makes the user frustrated and a tendency to reject the device will be developed. Robust myoelectric systems capable of adapting to these changes are very few. To achieve usability, more research initiatives have to be taken up to reduce the impact of these factors by adaptive algorithms. An effective adaptive system must know not only how to adapt but also when to adapt [55]. A clinically viable system demands adaptive algorithms which yield a completely stable, unsupervised solution.

Lack of sensory feedback other than sight is still another factor which hinders the usability of these current control systems. In physiological human motor control, once the commanded movements are generated, multimodal sensory feedback (e.g., vision, touch, proprioception) reaches the controller. This sensory-motor loop is very necessary for effective motor control. But in the case of amputees, the multimodal sensory feedback other than vision is not present. Most of the current MEC used today implemented only the feed forward or motor part of the loop which in turn made the system inefficient and unnatural.

Another significant factor that widens the gap between academia and industry state-of-the-art is that mostly PR-based systems are incapable of SPC of prostheses. Natural movements usually involve SPC of multiple DoFs. PR-based systems allow only sequential, on/off control where only a single function of the prosthetic device can be activated at a time. Also proportional

control cannot be obtained directly from classification. This makes the system quite unintuitive for prosthetic user. To realize this type of control, there are recent efforts in utilizing regression-based algorithms. Supervised and unsupervised algorithms have also been proposed and tested. Although works have been progressing on SPC, only marginal progress has been made in the clinically available solutions. Still bigger efforts are needed to address the problem. Following section reviews different approaches proposed for SPC of multi-functional prosthesis.

4. SIMULTANEOUS AND PROPORTIONAL CONTROL

Even though vast research have been carried out in MEC, it is not still matured enough to actuate all the functions provided by the prosthetic technology today. PR-based myoelectric techniques use a sequential strategy where only one class can be designated in one decision. Also, PR control which decides on an on-off basis, does not support proportional control of the prosthesis, crucial for optimizing the speed of its response. Contrary to this, natural limb movements involve continuous and simultaneous activation of multiple DoFs [48,49]. The methods used so far in the literature for SPC of multiple DoFs have been reviewed below. Table 4 provides a summary of these methods.

4.1 Method Based on Generative Model of sEMG

Jiang et al. proposed a method based on a biologically inspired generative model of the sEMG. The assumption made was that the muscles work in synergy, when the neural drives from the spinal level were shared by the motor unit. The neural drive produced corresponds to

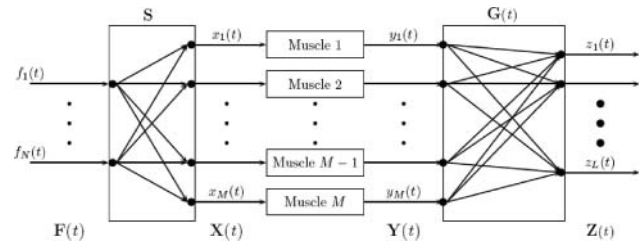


Figure 2: Biologically inspired generative model of the sEMG [56]

the different DoFs being activated when a natural movement occurs.

Figure 2 shows the schematic diagram of the generative model for sEMG, where $F(t)$ is the force function, S is an $M \times N$ matrix of activation coefficients, $X(t)$ is the activation function vector, M is the number of muscles active during a specific movement, $Y(t)$ is the sEMG produced from M , $G(t)$ is the volume conductor between muscles and electrodes, $Z(t)$ is the recorded sEMG [56]. Neural control information from sEMG was extracted by using a DoF wise nonnegative matrix factorization (NMF) [57]. A linear mapping between multichannel sEMG amplitude and force produced during wrist contraction involving flexion/extension, radial/ulnar deviation and the combination of both were provided [56]. The most important feature of the algorithm was its semi unsupervised nature. The force signals were not recorded; instead the DoF which was activated in specific contraction should be noted during training phase. Since the feature extracted, MSV and the operations used were simple matrix manipulation, the system would be computationally simple and could be easily implemented in real time.

R^2 index expresses the MLPs ability to estimate the force functions in each DoF. R^2 index for a training of the

Table 4: Methods for SPC

Refs.	Subjects	DoFs	Methodology	Features
[56]	12 N	Wrist F/E, R/U, P/S	DOF wise NMF	MSV
[58]	12 N	Wrist F/E, R/U, P/S	DOF wise NMF	Hudgin's TD
[59]	10 N, 1 CM	Wrist F/E, R/U	MLP with bilateral mirrored contraction	Hudgin's TD TDAR,TDWV
[60]	10 N, 1 CM	Wrist F/E, R/U	MLP with bilateral mirrored contraction	MSV, Hudgin's TD, TDAR, TDWV
[61]	6 N	Wrist F/E, R/U, P/S, Hand O/C	MLP with bilateral mirrored contraction	Envelopes of sEMG signals
[51]	6 N	Wrist F/E, R/U, P/S, Hand O/C	Hierarchy of LDA Classifiers	Hudgin's TD
[62]	5 N	Wrist rotation, F/E, Hand O/C	*Parallel dual-site control *PR control implemented using iEMG	MAV,TDAR
[63]	10 N	Wrist F/E, R/U, P/S	ANN with bilateral mirrored contraction	Hudgin's TD,
[64]	10 N, 1C	Wrist F/E, R/U	*Linear regression technique *Mixture of linear experts (ME) *MLP *Kernal ridge regression	Variance, RMS, Log-Var
[65]	10 N	Wrist rotation, F/E, Hand O/C	*Parallel dual-site control *Linear regression control	MAV of iEMG

CM – person having congenital malformation; N – normally limbed person.

MLP is defined as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^D \sum_{t=0}^N (f_i(t) - \hat{f}_i(t))^2}{\sum_{i=1}^D \sum_{t=0}^N (f_i(t) - \bar{f}_i(t))^2},$$

where D is the number of force functions, N is the number of data samples, $f_i(t)$ is the i th force function, $\hat{f}_i(t)$ is the corresponding estimate, and $\bar{f}_i(t)$ is the temporal average of $f_i(t)$. Total mean square error (MSE) of the estimates is given by the numerator of the equation and the total variance of the force functions by the denominator. Evaluation of R^2 index showed that the approach used in [56] had a poor performance in the case of pronation/supination.

4.2 Improved Method Based on Generative Model of sEMG

Nielsen et al. tried to improve the performance of the aforementioned method, by using Hudgin's TD features instead of MSV and by including data from combined DoF contractions in the training set instead of single DoF contractions [58]. The R^2 index increased to $95.1 \pm 1.2\%$ and $83.7 \pm 3.5\%$ for excluding and including DoF3, respectively. An improvement of 14% and 38.9%, respectively, were obtained on exclusion and inclusion of the third DoF. Both the studies discussed above investigated estimating the force function using the sEMG together with the measured forces from the wrist of the ipsilateral limb. But the approach could not be proven clinically, because the force function targets of the amputee's limb were no longer available in the case of unilateral upper-limb amputees.

4.3 Method Using Contra Lateral Limb

This issue was faced by Steffen et al. [59] by measuring the force generated at the contra lateral limb and detecting sEMG from the amputee limb. This was done under the assumption that the neuromuscular control information of the two upper limbs was similar while performing bilateral movements. Also performance of MLP-NN was conducted using additional feature sets. Better performance was shown by MLP when the feature sets; Hudgin's TD, TDAR coefficients and wavelet features concatenated with TD features referred to as TDWV feature sets were used. TD feature set improved the estimation performance significantly compared with others and was adequate since no improvement in performance could be shown by including AR or wavelet features.

The same strategy was used in [60] to develop an MES meant for unilateral amputees. MES from ipsilateral

limb were used to determine the intended isometric forces of the other limb's DoFs. This was made possible by an estimation algorithm trained using sEMG-force relation of the contralateral limb. The internal weights of the MLP were fixed during training, based on the difference between MLP output and actual forces. The R^2 index was determined during testing, by applying the best trained MLP output on a new test data set.

4.4 Method Based On Bilateral Mirrored Movements for Combined Activation Of Four DoFs

Though the studies in [59] and [60] offer a method for SPC in the case of unilateral amputees, it was limited to only two DOFs. Muceli et al. turned up with a better method in which an MLP is used to learn the association between sEMG and hand kinematics [61]. Contralateral hand kinematics during movements involving combined activation of four DOFs (included hand closing as the 4th DOF while the previous studies considered only wrist movements) was estimated from high density sEMG on the basis of bilateral mirrored movements. It was also proved that high density sEMG would allow to better exploiting the spatial information across the muscles compared with bipolar recordings routinely used in MEC studies. Accuracy in estimation was in the range 79% to 88% (R^2 index) for four DoFs in six able bodied subjects. Unlike previous studies the subjects hand was not restrained here, which was more related to real life scenarios. The method offered a more clinically viable means for SPC of multiple DoFs compared with the previous methods.

4.5 Hierarchy of LDA Classifiers

While single LDA classification strategy discriminates all discrete and combined classes as separate motions, parallel classification strategy discriminates each DoF individually using three LDA classifiers. Hierarchical classification strategy uses the output of DoFs higher in the hierarchy to choose classifiers for DoFs lower in the hierarchy. The hierarchical method was found to possess the most promising classification error below 15% on average for discrete and combined motion. But this study was limited by the use of healthy control subject and lack of online performance measures. Each box in Figure 3 is an LDA classifier with motion classes from one or more DoFs.

4.6 Method Using Parallel Dual-Site Control

Smith et al. suggested that SPC of a multi DoF prosthetic wrist hand system can be provided by using iEMG along with a parallel dual-site control paradigm [62]. DoFs

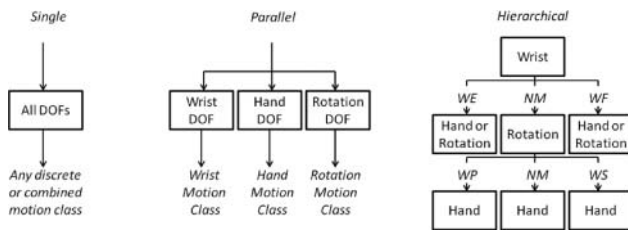


Figure 3: Block diagrams describing examples of each control strategy for a 3-DoF controller [51]

were controlled by the in iEMG from a physiologically appropriate antagonistic muscle pair. Since cross talk was minimal in iEMG, MES from multiple muscle pairs could be used in parallel to simultaneous control of multiple DoFs. Thus, the potential of parallel dual-site control was hypothesized.

4.7 Constrained and Unconstrained-Based Control Method

Ameri et al. made a comparison of three control paradigms of constrained (force targeted), unconstrained (position targeted) and resisted unconstrained (position targeted) limb contractions. They were compared offline using estimation accuracies (R^2) and online using a real-time computer-based target acquisition test [63]. As in the previous cases mirrored bilateral contractions were used to train ANN for simultaneous MEC of three DoFs. The constrained control model outdid the unconstrained control model in the abduction–adduction DoF (R^2 constrained = 90.8 ± 0.6 , R^2 unconstrained = 85.6 ± 1.6) and pronation–supination DoF (R^2 constrained = 88.5 ± 0.9 , R^2 unconstrained = 82.3 ± 1.6), but no significant difference was found in the flexion–extension DoF. Also, in two real-time testing metrics including completion time and path efficiency, constrained control method exhibited better performance. The results of constrained method, however, were almost similar to that of the resisted unconstrained method (with braces) in both offline and real-time tests. This suggested that when comparable levels of muscle activation was used, the quality of control using constrained and unconstrained contraction based myoelectric schemes was not substantially different, i.e., when similar sEMG levels are used the two protocols showed similar control performance.

4.8 Linear and Non-Linear Regression Method

Hahne et al. made a comprehensive comparison of the linear and nonlinear regression methods used in independent, proportional and simultaneous myoelectric control of multiple DoFs [64]. Kernel ridge regression

(KRR) technique was found to have a better performance than linear regression, Mixture of linear experts (MEs) and MLP. It was also found that a logarithmic transformation of the well-established variance feature linearized the relation between sEMG and wrist angle. This permitted the application of less complex and computationally inexpensive linear methods. Training data need not be recorded for all possible combinations of DoFs. ME, a physiologically inspired extension of linear regression, represented a promising candidate for the future prosthetics.

4.9 Parallel Dual-Site and Linear Regression Control

In [65] a real-time performance of the two main techniques, parallel dual-site and linear regression control, used for SPC were implemented by using iEMG and investigations were carried out. The type of training data significantly influenced the offline prediction accuracy for different DoF-complexities. Training with 1-, 2-, and 3-DoF data provided the best prediction of 3-DoF movements ($R^2 = 0.74 \pm 0.0$), whereas training with only 1-DoF data produced significantly worse predictions ($R^2 = 0.51 \pm 0.03$; $p < 0.001$). In contrast, training with only 1-DoF data allowed for better predictions of 1-DoF movements ($R^2 = 0.75 \pm 0.03$), at the expense of poorer performance in predicting 3-DoF movements ($R^2 = 0.72 \pm 0.03$; $p < 0.001$). It has been hypothesized that 1-DoF movements can be easily isolated using dual-site control. However, linear regression control exhibited better performance when acquiring targets requiring use of all three DoFs. Multiple DoFs could be activated simultaneously, but at the cost of unintended movements when trying to isolate individual DoFs.

5. CONCLUSION

Upper-arm amputation can dramatically affect the capabilities of a person. An estimate of three million people was living with a major loss of an upper limb in 2009. sEMG has been used as the control for myoelectric upper-limb prosthesis since 1960s. Despite several years of academic research in PR-based control which yielded excellent classification performance in laboratory implementation, commercial devices are still based on amplitude coding approach proposed some 60 years ago. The challenges that need to be addressed before successful commercial implementation of a robust system include lack of SPC, factors of variability such as change in arm posture, muscle contraction level, electrode repositioning and fatigue which degrades control, absence of sensory input beyond just visual feedback and lack of

multimodal approaches which could lead to the development of intelligent controllers. Minimum or absence of training and limited computational cost are other highly desirable features.

This paper focused on different control strategies available in the literature for SPC of multiple DoFs which is the most significant among the open issues. Regression approaches based on NMF, ANN or LR and modification of classical PR paradigm have been identified as promising approaches for SPC. However, the potential of regression approaches in SPC is found to be superior when compared with others. Better control may be achieved by utilizing the best features of PR algorithm in the less explored regression algorithms. Robust and intuitive myoelectric prostheses can one day become a reality in the everyday life of amputees but the path still requires many more practical steps.

DISCLOSURE STATEMENT

No potential conflict of interest was reported by the authors.

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Authors



Nisheena V. Iqbal is currently pursuing her PhD degree in electronics and communication engineering at Karpagam University, Coimbatore, India, and working as associate professor in MES College of Engineering, Kuttippuram, India. Her research interests include biomedical signal processing.

E-mail: nisheenasagil@gmail.com



Shaniba Asmi P. is currently pursuing her PhD degree in electronics and communication engineering at Karpagam University, Coimbatore, India, and working as an associate professor in MES College of Engineering, Kuttippuram, India. Her research interests include biomedical signal processing.

E-mail: shanibasmi@gmail.com



Kamalraj Subramaniam received his PhD degree in mechatronic engineering from University Malaysia Perlis, Perlis, Malaysia in 2014. Currently, he is an associate professor and deputy head (human machine interface cluster) Karpagam University, Coimbatore, India. His research interests include biomedical signal processing, artificial neural networks, VLSI design.

E-mail: kamalrajece@gmail.com
